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Diagnosis information - Reversing camera

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Diagnosis information - Reversing camera control unit

Button Reversing camera

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Short circuit to B+

Fault setting condition(s)

- Voltage internal : > 6 V

Fault effects

- ♦ No function

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **-0.5 V** ⇒ **+0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance > **0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance < **2 Ohm**

Diagnosis information - Reversing camera control unit

Indicator light Reversing camera

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Indicator light** on

Possible fault causes

- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Open circuit

Fault setting condition(s)

- Voltage internal : < -4,0 V (Short circuit to ground)
- Voltage internal : > +4,0 V (Short circuit to B+)
- Voltage internal : > +3,5 V (Open circuit)

Fault effects

- ♦ No function at Short circuit to ground and Open circuit

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **-0.5 V** ⇒ **+0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

◆ Setpoint resistance < **2 Ohm**

Diagnosis information - Reversing camera control unit

Indicator light Reversing camera

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Indicator light** on

Possible fault causes

- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Open circuit

Fault setting condition(s)

- Voltage internal : < -4,0 V (Short circuit to ground)
- Voltage internal : > +4,0 V (Short circuit to B+)
- Voltage internal : > +3,5 V (Open circuit)

Fault effects

- ♦ No function at Short circuit to ground and Open circuit

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **-0.5 V** ⇒ **+0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

◆ Setpoint resistance < **2 Ohm**

Diagnosis information - Reversing camera control unit

Indicator light Reversing camera

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Indicator light** on

Possible fault causes

- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Open circuit

Fault setting condition(s)

- Voltage internal : < -4,0 V (Short circuit to ground)
- Voltage internal : > +4,0 V (Short circuit to B+)
- Voltage internal : > +3,5 V (Open circuit)

Fault effects

- ♦ No function at Short circuit to ground and Open circuit

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **-0.5 V** ⇒ **+0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

◆ Setpoint resistance < **2 Ohm**

Diagnosis information - Reversing camera control unit

No video signal for PCM

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- PCM on

Fault setting condition(s)

- Short circuit to ground (Voltage internal : < 0,25 V)
- Short circuit to B+ (Voltage internal : > 1,9 V)
- Open circuit (Voltage internal : 0,725 V to 1,9 V)

Fault effects

- ♦ No function

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Diagnosis information - Reversing camera control unit

No video signal for PCM

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- PCM on

Fault setting condition(s)

- Short circuit to ground (Voltage internal : < 0,25 V)
- Short circuit to B+ (Voltage internal : > 1,9 V)
- Open circuit (Voltage internal : 0,725 V to 1,9 V)

Fault effects

- ♦ No function

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Diagnosis information - Reversing camera control unit

No video signal for PCM

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- PCM on

Fault setting condition(s)

- Short circuit to ground (Voltage internal : < 0,25 V)
- Short circuit to B+ (Voltage internal : > 1,9 V)
- Open circuit (Voltage internal : 0,725 V to 1,9 V)

Fault effects

- ♦ No function

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Diagnosis information - Reversing camera control unit

Camera left
Camera right
Camera rear

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Reverse gear or transmission range R selected
- PCM on

Fault setting condition(s)

- Short circuit to ground (Voltage: < 0,46 V)
- Short circuit to B+ (Voltage: > 1,86 V)
- Open circuit
- Implausible signal

Fault effects

- ♦ No function

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

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Camera left
Camera right
Camera rear

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Fault setting condition(s)

- Short circuit to ground (Voltage: < 0,46 V)
- Short circuit to B+ (Voltage: > 1,86 V)
- Open circuit
- Implausible signal

Fault effects

- ♦ No function

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

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Fault setting condition(s)

- Short circuit to ground (Voltage: < 0,46 V)
- Short circuit to B+ (Voltage: > 1,86 V)
- Open circuit
- Implausible signal

Fault effects

- ♦ No function

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

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Diagnosis information - Reversing camera control unit

Camera left
Camera right
Camera rear

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Reverse gear or transmission range R selected
- PCM on

Fault setting condition(s)

- Short circuit to ground (Voltage: < 0,46 V)
- Short circuit to B+ (Voltage: > 1,86 V)
- Open circuit
- Implausible signal

Fault effects

- ♦ No function

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Diagnosis information - Reversing camera control unit

Supply voltage Camera

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Supply voltage < 4 V
- Supply voltage > 7,5 V

Possible fault causes

- ♦ Wiring harness
- ♦ Reversing camera control unit

Fault effects

- ♦ No function

Diagnosis information - Reversing camera control unit

Supply voltage Camera

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Supply voltage < 4 V
- Supply voltage > 7,5 V

Possible fault causes

- ♦ Wiring harness
- ♦ Reversing camera control unit

Fault effects

- ♦ No function

Diagnosis information - Reversing camera control unit

Activation line to PCM

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground : Voltage internal < (internal monitoring voltage - 5,7 V)
- Short circuit to B+ : Voltage internal > (internal monitoring voltage + 0,4 V) with Activation Active or Voltage internal > 6 V at Activation not active
- Open circuit: internal monitoring voltage - 1,48 V to + 0,4 V

Fault effects

- ♦ none Activation of PCM

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Diagnosis information - Reversing camera control unit

Activation line to PCM

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground : Voltage internal < (internal monitoring voltage - 5,7 V)
- Short circuit to B+ : Voltage internal > (internal monitoring voltage + 0,4 V) with Activation Active or Voltage internal > 6 V at Activation not active
- Open circuit: internal monitoring voltage - 1,48 V to + 0,4 V

Fault effects

- ♦ none Activation of PCM

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**